

Two Qubit Reduction sample with UCCSD fails with missing Z2_symmetric

<https://github.com/qiskit-community/qiskit-nature/issues/153>

ROW 86

Two Qubit Reduction sample with UCCSD fails with missing Z2_symmetries argument #153

New issue

Closed



woodsp-ibm opened on Apr 23, 2021 · edited by woodsp-ibm

Edits ...

The following error results when running the code sample that follows. Run with `two_qubit_reduction=False` on the converter and it works fine. I think the problem is more likely in the code in Terra, but since most of the sample is nature I raised it here but it can be transferred to Terra if it indeed turns out to be where the problem needs fixing.

```
File "qiskit_nature/circuit/library/ansatzes/evolved_operator_ansatz.py", line 165, in <listcomp>
    evolved_ops = [self.evolution.convert((coeff * op).exp_i()) for op in self.operators]
File "qiskit/opflow/operator_base.py", line 355, in __rmul__
    return self.mul(other)
File "qiskit/opflow/primitive_ops/pauli_sum_op.py", line 138, in mul
    return super().mul(scalar)
File "qiskit/opflow/primitive_ops/primitive_op.py", line 138, in mul
    return self.__class__(self.primitive, coeff=self.coeff * scalar)
TypeError: __init__() missing 1 required positional argument: 'z2_symmetries'
```

Here is the code sample to reproduce the above:

```
from qiskit_nature.drivers import PySCFDriver, UnitsType
from qiskit_nature.problems.second_quantization.electronic import ElectronicStructureProblem
# Use PySCF, a classical computational chemistry software
# package, to compute the one-body and two-body integrals in
# electronic-orbital basis, necessary to form the Fermionic operator
driver = PySCFDriver(atom='H .0 .0 .0; H .0 .0 0.735',
                    unit=UnitsType.ANGSTROM,
                    basis='sto3g')

problem = ElectronicStructureProblem(driver)
# generate the second-quantized operators
second_q_ops = problem.second_q_ops()
main_op = second_q_ops[0]
num_particles = (problem.molecule_data.transformed.num_alpha,
                problem.molecule_data.transformed.num_beta)
num_spin_orbitals = 2 * problem.molecule_data.num_molecular_orbitals
# setup the classical optimizer for VQE
from qiskit.algorithms.optimizers import L_BFGS_B
optimizer = L_BFGS_B()
# setup the mapper and qubit converter
from qiskit_nature.mappers.second_quantization import ParityMapper
from qiskit_nature.converters.second_quantization import QubitConverter
mapper = ParityMapper()
converter = QubitConverter(mapper=mapper, two_qubit_reduction=True)
# map to qubit operators
qubit_op = converter.convert(main_op, num_particles=num_particles)
# setup the initial state for the ansatz
from qiskit_nature.circuit.library import HartreeFock
init_state = HartreeFock(num_spin_orbitals, num_particles, converter)
# setup the ansatz for VQE
from qiskit_nature.circuit.library import UCCSD
ansatz = UCCSD(converter, num_particles, num_spin_orbitals, 1, init_state)
from qiskit import Aer
backend = Aer.get_backend('statevector_simulator')
# setup and run VQE
from qiskit.algorithms import VQE
algorithm = VQE(ansatz,
                optimizer=optimizer,
                quantum_instance=backend)
result = algorithm.compute_minimum_eigenvalue(qubit_op)
print(result.eigenvalue.real)
```

The issue was originally raised on the external Slack channel:

<https://qiskit.slack.com/archives/CB6C24TPB/p1619192806297800>



woodsp-ibm added type: bug on Apr 23, 2021

woodsp-ibm assigned ikkohan on Apr 23, 2021

Assignees

ikkohan

Labels

type: bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



```
# setup the mapper and qubit converter
from qiskit_nature.mappers.second_quantization import ParityMapper
from qiskit_nature.converters.second_quantization import QubitConverter
mapper = ParityMapper()
converter = QubitConverter(mapper=mapper, two_qubit_reduction=True)
# map to qubit operators
qubit_op = converter.convert(main_op, num_particles=num_particles)
# setup the initial state for the ansatz
from qiskit_nature.circuit.library import HartreeFock
init_state = HartreeFock(num_spin_orbitals, num_particles, converter)
# setup the ansatz for VQE
from qiskit_nature.circuit.library import UCCSD
ansatz = UCCSD(converter, num_particles, num_spin_orbitals, 1, init_state)
from qiskit import Aer
backend = Aer.get_backend('statevector_simulator')
# setup and run VQE
from qiskit.algorithms import VQE
algorithm = VQE(ansatz,
                optimizer=optimizer,
                quantum_instance=backend)
result = algorithm.compute_minimum_eigenvalue(qubit_op)
print(result.eigenvalue.real)
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woodsp-ibm added **type: bug** on Apr 23, 2021

woodsp-ibm assigned [ikkoham](#) on Apr 23, 2021



mrossinek on Apr 23, 2021

Member ...

Seems to be a duplicate of this issue: [Qiskit/qiskit#6127](#)



mrossinek on Apr 28, 2021

Member ...

Once [Qiskit/qiskit#6315](#) is merged we need to remove [this line](#) in order to close this issue.
While validating the fix of the PR in Terra I noticed that we can probably also remove the `@slow_test` decorators because these tests aren't actually that slow..



mrossinek on May 5, 2021

Member ...

Closed by [Qiskit/qiskit#6315](#)



mrossinek closed this as **completed** on May 5, 2021

mrossinek mentioned this on May 5, 2021

[Re-enable TwoQubitReduction + UCCSD test #170](#)

Re-enable TwoQubitReduction + UCCSD test #170

<> Code

Merged mrossinek merged 1 commit into `qiskit-community:main` from `mrossinek:re-enable-two-q-reduction-tests` on May 5, 2021

Conversation 0 Commits 1 Checks 0 Files changed 1

+0 -1



mrossinek commented on May 5, 2021

Member

Summary

Re-enables the TwoQubitReduction + UCCSD test case after closing [#153](#)
(Technically, this should close that issue, but I closed it prematurely...)

Details and comments



Re-enable TwoQubitReduction + UCCSD test **Verified** 35888af

mrossinek added the `run_slow` label on May 5, 2021

mrossinek requested review from [manoelmarques](#), [pbark](#), [stefan-woerner](#) and [woodsp-ibm](#) as code owners
4 years ago



woodsp-ibm approved these changes on May 5, 2021 [View reviewed changes](#)

mrossinek merged commit `64bdc92` into `qiskit-community:main` on May 5, 2021

mrossinek deleted the `re-enable-two-q-reduction-tests` branch 4 years ago

Anthony-Gandon pushed a commit to Anthony-Gandon/qiskit-nature that referenced this pull request on May 25, 2023

Re-enable TwoQubitReduction + UCCSD test (`qiskit-community#170`) **Unverified** bacf551

Reviewers

woodsp-ibm	✓
manoelmarques	•
pbark	•
stefan-woerner	•

Assignees

No one assigned

Labels

`run_slow`

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

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2 participants